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Large Language Models, Small Labor Market Effects

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Abstract

We examine the early labor market impacts of AI chatbots by linking large-scale, representative adoption surveys to administrative labor records in Denmark. Using difference-in-differences, we estimate precise null effects on earnings and recorded hours at both the worker and workplace levels, ruling out effects larger than 2% two years after. These null results hold for intensive users, early adopters, workplaces with substantial investments, workers reporting large gains, flexible-pay occupations, and early-career jobs. Adoption is linked to occupational switching and task restructuring, but without net changes in hours or earnings. Our findings challenge narratives of imminent disruption from Generative AI.

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for workers’ pre-determined characteristics—including gender, age, and labor market experience—to ensure these factors do not drive our estimates (e.g., adopters being younger and naturally on upward earnings trajectories). Second, we leverage our panel data to implement a difference-in-differences approach indexed to November 2022, the release date of ChatGPT. This allows us to control for time-invariant differences between workers (e.g., high-ability adopters who would have earned more regardless) and examine whether adopters experienced differential changes after this date. Additionally, we assess whether adopters were on distinct labor market trends even before AI chatbots became available, enabling us to control for potential pre-existing differences.

A third question is timing: When should we date the arrival of AI chatbots, and when should we expect their effects on labor market outcomes to appear? We use the launch of ChatGPT in November 2022 as a before-and-after moment in an event-study design, marking a sharp rise in public awareness of AI chatbots.⁸ We also separately examine effects for early adopters—workers observed using chatbots in our 2023 survey round. Still, even if adoption occurs rapidly, labor market effects may take time to materialize. We examine these adjustment dynamics in two ways. First, we implement a dynamic difference-in-differences design to trace how effects unfold monthly following the introduction of AI chatbots. Second, we analyze heterogeneity across occupations that vary in the flexibility of labor rules.

A final concern is that difference-in-differences identifies only *differential* effects for adopters—the “missing intercept.” Appendix F explores potential spillovers to non-adopters from several angles, all of which suggest that AI chatbots have had no meaningful impact on their labor market outcomes to date.

⁸This timing is motivated by the rapid uptake of ChatGPT among workers: most adopters began using it within a year of its launch (Humlum and Vestergaard, 2025). Appendix E.1.3 shows that ChatGPT remains the dominant AI chatbot to date. Moreover, event studies centered on its launch in November 2022 have become common in the literature (e.g., Brynjolfsson, Chandar and Chen (2025); Eisfeldt et al. (2024); Lichtinger and Hosseini Maasoum (2025); Schubert (2025); Teutloff et al. (2025)), facilitating comparison across studies.